

Knowing the Partner's Objective Increases Embodiment towards a Limb Controlled by the Partner

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ABSTRACT

We have developed a joint avatar of which two participants simultaneously control left and right-side limbs in first-person view in a virtual environment. We aimed to investigate whether having a common objective with a partner affects sense of embodiment towards a limb controlled by the partner. Participants performed reaching tasks using the joint avatar. We found that the embodiment towards the arm controlled by the partner was significantly higher when the participant dyads shared a common objective or when they were allowed to see their partner's goal, compared to when their partner's goal was unknown to them.

Keywords: Joint avatar, Sense of embodiment, Common objective

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Humans can have illusory body ownership (embodiment) to non-human objects and virtual avatars using visual-tactile integration and visuo-motor synchronicity [1,2]. Using virtual reality (VR), it is possible to go beyond the constraint of “one body for one person”, allowing multiple users to share one avatar to perform complicated tasks. Even though a few studies on embodiment of shared avatars (which are avatars controlled by two individuals where the movement of one avatar is determined by averaging two individual's motion) have been reported [3,4], there are no reported studies on embodiment towards body parts completely controlled by a partner. Such jointly controlled avatars (joint avatars) or robots can be utilized to combine strengths of two or more people to increase efficiency in multi-tasking, aid disabled VR users or to simply share different limbs of one avatar with other online users during VR games/experiences.

However, for enhancing embodiment towards such shared avatars, identifying the factors that affect/limit embodiment is crucial. In this study, we created a joint avatar of which left and right-side limbs could be controlled by two individuals. We conducted an experiment where participant dyads performed three types of reaching tasks using both hands of the “Joint avatar” (Figure 1).

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When two individuals act together (joint action), knowing what the partner perceives and what the goal is (joint attention), is important [5]. Therefore, we hypothesized that the embodiment towards the arm controlled by the partner would be improved when the participants share one goal and/or know the partner's goal. To test this, we investigated whether the sense of embodiment would be higher with one identical goal than two different goals to be reached by left and right limbs of the joint avatar, and whether the sense of embodiment would be higher when the partner's goal was visible compared to when it was invisible using two different goals.



Figure 1: Joint avatar

2 METHODS

2.1 Participants

Twenty university students as ten dyads (All males, mean age 22.3, SD 2.4) participated in the experiment as dyads. All participants were unaware of the hypothesis of the experiments and had normal or corrected-to-normal eyesight. They provided informed consent before the experiment. The methods of the experiment were approved by the Ethical Committee for Human-Subject Research at the Toyohashi University of Technology.

2.2 Apparatus

The movements of the two participants were measured by a motion-capture system (Vicon Bonita10, 12 cameras, 1024 x 1027 pixels, 250 fps). Two computers with the same specifications received the processed information of the motion capture data and presented the virtual environment on head-mounted displays (HMD: HTC VIVE Pro Eye, 1440 x 1600 pixels per eye, 90 x 110deg, 90 Hz refresh) of both participants. The virtual environment and experiment tasks were created using Unity (2017.4.1f1). BIOPAC MP 160 and wireless PPG and EDA amplifiers (BN-PPGED) were used to record the skin conductance response (SCR). A wireless transmitter (BN-PPGED) was placed on each participant's wrist. Two disposable electrodes (EL507) were pasted to the distal phalanges of their middle and ring fingers. Two lead cables (BN-EDA-LEAD2) connected the wireless transmitter to the electrodes. SCR data were recorded at 1000 Hz.

2.3 Stimuli and Conditions

The experiment consisted of three conditions. In each session, one participant controlled the left side of the joint avatar while the other controlled the right side throughout the experiment, both in first-person view. We named the three conditions “Same goal”, “Different goals: visible”, and “Different goals: invisible”. In the “same goal condition”, one yellow target appeared in front of the joint avatar and participants were asked to collaborate and touch the target with both hands of the joint avatar. In the different goals: visible condition, two targets (one blue and one red) appeared in front of the avatar and participants were asked to touch the blue target with the right hand and red target with the left hand of the joint avatar. Different goals: invisible condition was the same as different goals: visible condition, but the target of their partner was invisible for each participant. All targets were spheres of 5cm diameter and were erased 200ms after reached appropriately for each condition. After erasing, the next target appeared 2s later. Each session consisted of 100 reaches per participant and on the final reach of each session, a stimulus of a knife stabbing either the left or the right hand of the joint avatar was displayed.

2.4 Procedure

The experiment consisted of six sessions (3 conditions x 2 repetitions). In the two repetitions for each condition, left hand was stabbed at the end of one repetition while the right hand was stabbed in the other. The order of sessions was randomized for each dyad. One session lasted 5-7 minutes and after each session, both participants answered a questionnaire regarding how they felt about the left and right arms of the joint avatar. The questionnaire was based on a previous study [6]. Using the answers, we calculated senses of ownership and agency for the “controlled arm” (the arm of the joint avatar controlled by the participant), and the “non-controlled arm” (arm of the joint avatar controlled by the partner of the participant). Skin conductance response was calculated as the difference between the maximum value detected in a 10s post-stimulus (knife) time window and the baseline calculated as the average value of a 5s pre-stimulus time window (for both participants in each session).

3 RESULTS

Two-way repeated measures ANOVA with ART (aligned rank transformation) procedure [7] was conducted for ownership (Figure 2), agency (Figure 3), and SCR data. There were significant interactions between the condition and arm in both senses of ownership and agency (Ownership: $F(2,38)=7.622$, $p=0.002$, $\eta_p^2=0.286$, Agency: $F(2,38)=8.141$, $p=0.001$, $\eta_p^2=0.300$). Significant simple main effects on the goal condition were seen only in the non-controlled side (Ownership: $F(2,38)=9.292$, $p<0.001$, $\eta_p^2=0.328$, Agency: $F(2,38)=0.211$, $p=0.011$, $\eta_p^2=0.211$). Tukey post hoc tests confirmed that both ownership and agency were significantly higher in the same goal condition compared to different goals: invisible condition (Ownership: $p=.0015$, Agency: $p=.0181$) and significantly higher in the different goals: visible condition compared to different goals: invisible condition (Ownership: $p=0.0021$, Agency: $p=0.0308$). The SCR was higher on the controlled arm than the non-controlled arm ($F(1,19)=6.8201$, $p=0.017$, $\eta_p^2=0.264$), but not different between conditions of goals.

4 DISCUSSION

There were no significant differences of ownership or agency towards the non-controlled arm among the same goal and different goal: visible conditions. However, ownership and agency towards the non-controlled arm were higher (less negative) in both same goal and different goals: visible conditions compared to the different goals: invisible condition. This suggests that being able to

know the partner’s objective increases embodiment towards the arm controlled by the partner regardless of the shared or divided nature of the task goal. However, since it was not supported by SCR data, further investigation of other physiological measures is necessary in future work.

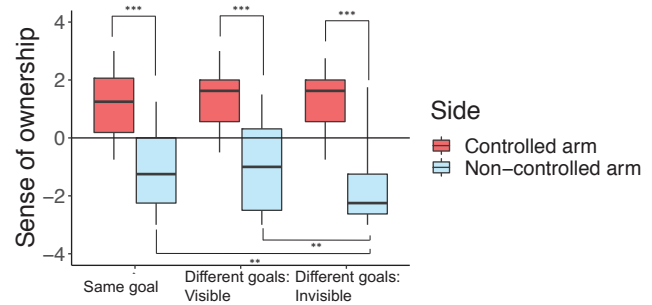


Figure 2: Results of sense of ownership

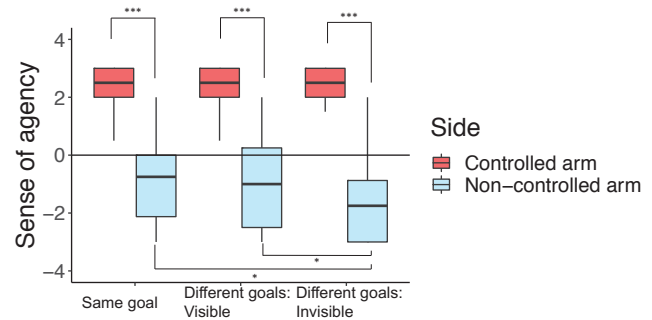


Figure 3: Results of sense of agency

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